Bar Graphs to go with Within-Groups ANOVA

Analyze → Graphs → Legacy Dialogs → Bar

As for the WG ANOVA…

- Each quantitative variable (clark_age & lois_age) holds the DV for one of the IV conditions

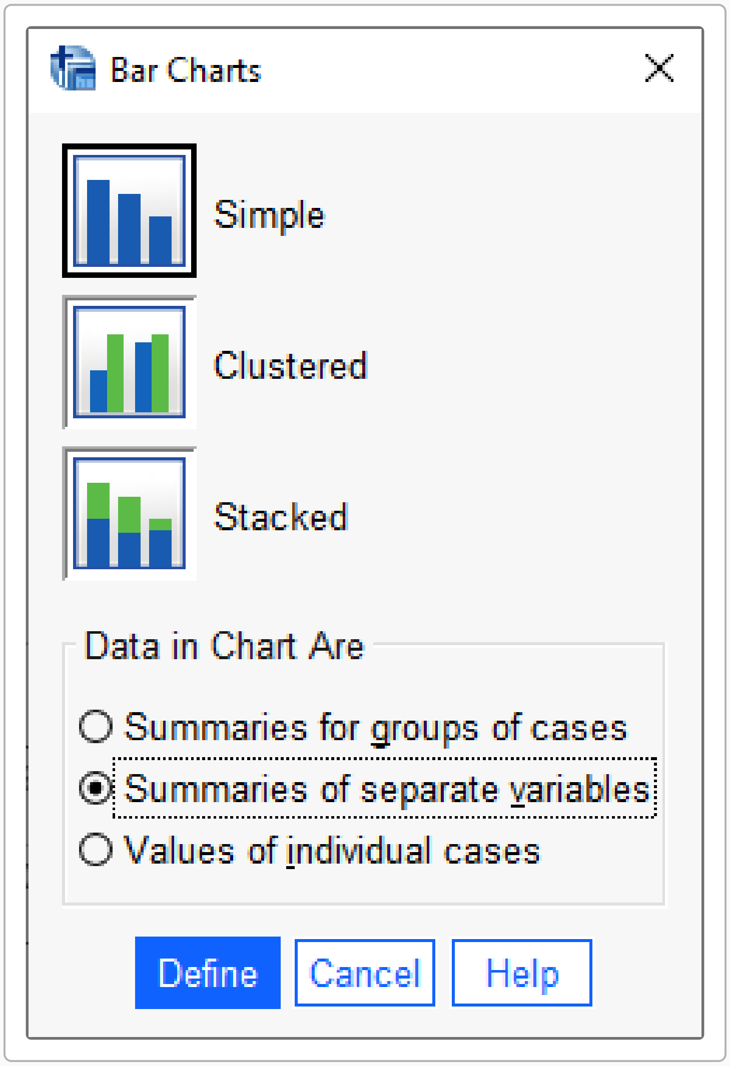
#1 Select type of chart

- Click on "Simple" icon
- Click the "**Summaries of separate variables**" radio button
- Click "Define"

For within-groups designs, use "Summaries of separate variables" — **not** "Summaries for groups of cases" (which is for between-groups designs).

#2 Select the Variables

- Highlight the variables
- Click the arrow to move the DV into the "Variable" window

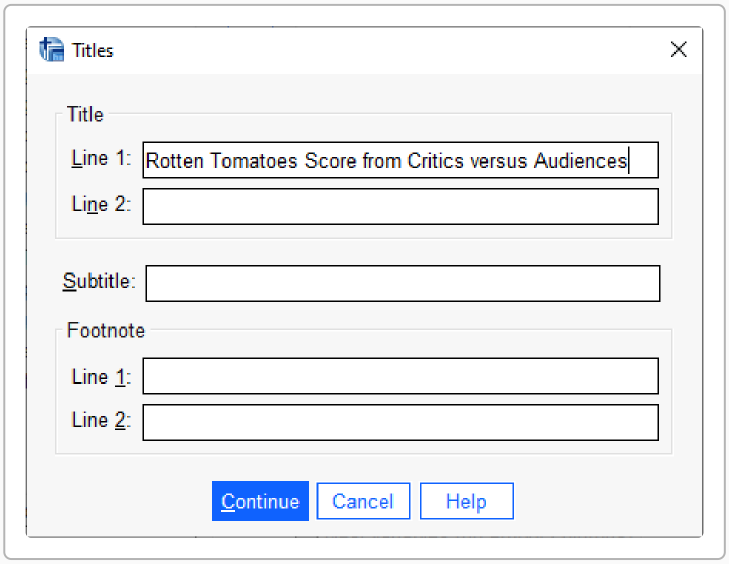


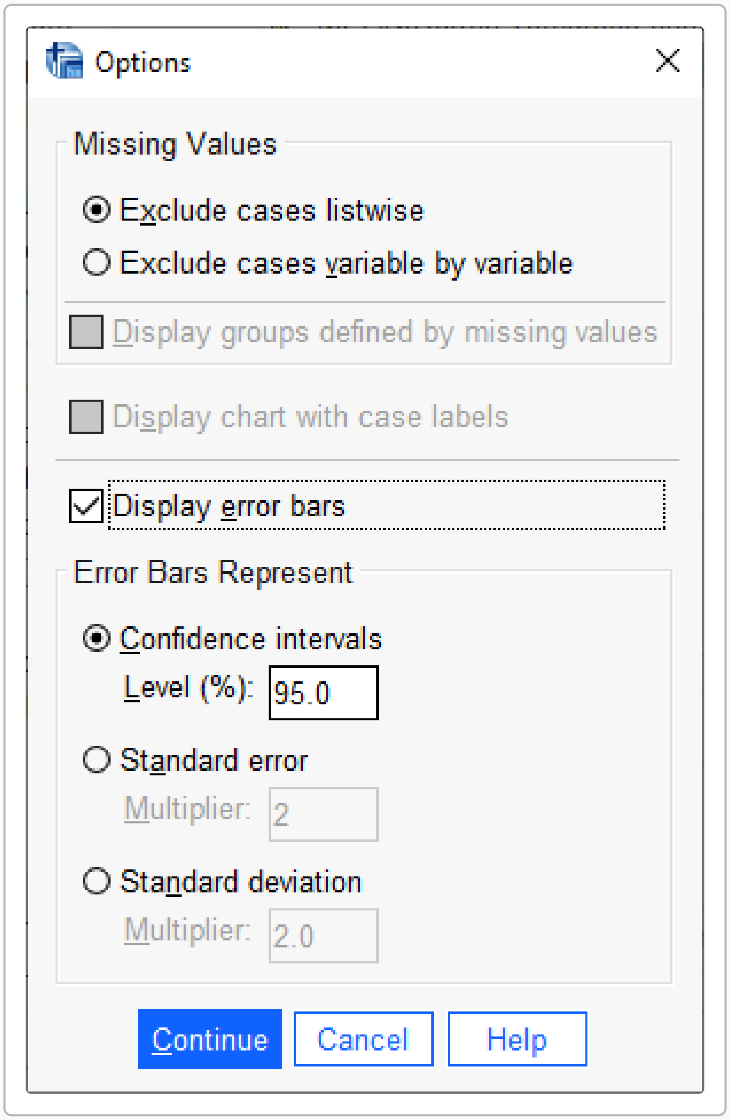
#3 Write the Title

- Click on "Titles"
- Type titles, subtitles & footnotes as desired
- Click "Continue"

#4 Select the Whiskers

- Click Options
- Check "Display error bars"
- Check "Standard Error" or "Confidence intervals"
- Click Continue


Titles
Title
Line 1: Rotten Tomatoes Score from Critics versus Audiences
Line 2:
Subtitle:
Footnote
Line 1:
Line 2:
Continue
Cancel
Help


Options
Missing Values
Exclude cases listwise
Exclude cases variable by variable
Display groups defined by missing values
Display chart with case labels
Display error bars
Error Bars Represent
Confidence intervals
Level (%): 95.0
Standard error
Multiplier: 2
Standard deviation
Multiplier: 2.0
Continue
Cancel
Help

SPSS Output

Interpreting the Graph

For within-groups designs, each bar represents the **mean** of one condition. In this example:

- The **Clark Kent bar** shows the mean age of actors who have played Clark Kent
- The **Lois Lane bar** shows the mean age of actors who have played Lois Lane

The **error bars** (vertical lines extending from each bar) show the **variability within each condition** — specifically, the 95% confidence interval around each mean. Wider error bars indicate more variability in the data, while narrower bars indicate consistency.

#5 Using the graph

Right-click the graph in the SPSS output window and select "Copy" to put the graph into a Word or other file.